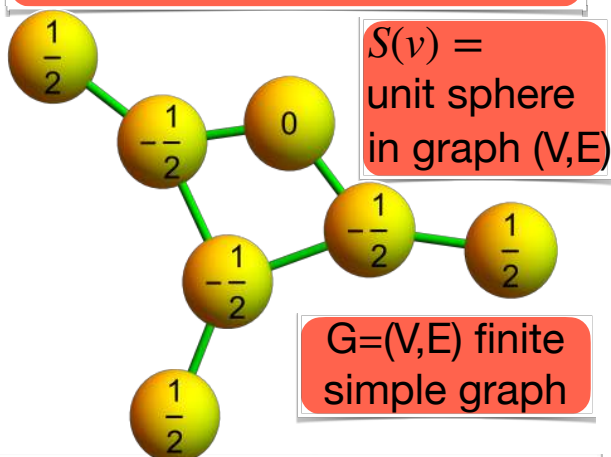


Gauss-Bonnet

$$\sum_{x \in V} K(v) = \chi(M)$$

$$\chi(G) = \sum_k (-1)^k f_k(G) \quad \text{Euler char.}$$

$$K(v) = 1 - \sum_{k=0}^{q-1} (-1)^k \frac{f_k(S(v))}{k+1}$$



$$f_k(A) = \# \text{ of } k - \text{simplices in } A$$

More general:

$$f'_G(t) = \sum_{v \in V} f_{S(v)}(t)$$

$$\text{simplex generating function} \\ f(t) = 1 + f_0 t + f_1 t^2 + \dots + f_q t^{q+1}$$

$$S(v) = \{w \in V, (v, w) \in E\}$$

unit sphere of a vertex v

Example:

7 vertices 7 edges

$$f_G(t) = 1 + 7t + 7t^2$$

$$f'_G(t) = 7 + 14t$$

$$f(-1) = \chi(G)$$

simplex
generating
functions
of $S(v)$ add
up to $f'(t)$

$$\begin{bmatrix} 3t+1 \\ 2t+1 \\ 3t+1 \\ 3t+1 \\ t+1 \\ t+1 \\ t+1 \end{bmatrix} \quad \text{functional curvatures}$$

Origin

Descartes (1623)

De solidorum elementis

$$\alpha(v) = 2\pi(1 - \frac{d(v)}{6})$$

Eberhard (1891)

Morphologie der Polyheder

$$\sum_k (6-k)p_k = 12$$

Heawood (1890)

Discharge in graph coloring

$$6 - d(v)$$

Levitt (1992):

for simplicial complexes

$$K(v)$$

$$K_2 = 1 - E/6$$

Knill (2012):

rediscover Levitt's curvature
graph set-up

$$K_4 = 1 - E/6 + F/10$$

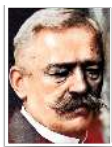
small dim. manifolds

Knill (2019):

Dehn-Sommerville story

$$f'_G(t) = \sum_{v \in V} f_{S(v)}(t)$$

Eberhard



Heawood



Levitt



Continuum

$\alpha, \beta, \gamma = \text{flat angles}$

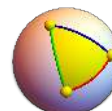
$$\pi = \alpha + \beta + \gamma$$



Euclid

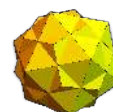
$\alpha, \beta, \gamma = \text{spherical angles}$

$$\pi + A = \alpha + \beta + \gamma$$



Harriot

$$\sum_i \kappa_i = 4\pi$$



$\kappa_i = \text{excess angle}$

Descartes

$$\int_0^L \kappa_g(t) dt = 2\pi$$



Hopf

$$\kappa_g = \dot{x} \times \ddot{x}$$

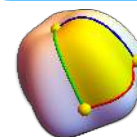
$$K = \det(A) = \frac{\det(II)}{\det(I)}$$

$$\kappa_g = (\dot{x} \times n) \cdot \ddot{x}$$

Gauss

$$\int_U K dV + \sum_i \int_{C_i} \kappa_g dt + \sum_i \kappa_i = 2\pi$$

$\kappa_i = \text{complementary angle}$



$$\int_U K dV = 2\pi \chi(M)$$



Bonnet

dk Gauss Bonnet Chern integrand

M compact $2k$ -dim Riemannian manifold

$$\int_M K dV = \chi(M)$$



Chern